

Courier Management System (CMS)

User Story:

The Courier Management System (CMS) is designed to manage the complete process of courier booking, shipment tracking, delivery, and customer management in an efficient and organized way. Customers can register and log in to the system, book courier services, enter sender and receiver details, make payments, and track their packages using a unique tracking number. Delivery agents can view their assigned shipments, update the delivery status, and confirm successful deliveries. Administrators can manage customers, shipments, delivery agents, payments, and overall courier operations. The system helps reduce manual work, improves delivery tracking, provides real-time status updates, and makes the courier management process faster, easier, and more reliable for customers and staff.

1. INTRODUCTION

1.1 Project Overview :

The Courier Management System (CMS) is a software application developed to manage and automate the various activities involved in courier and parcel delivery services. The system allows customers to register, book courier services, provide sender and receiver details, make payments, and track their parcels using a unique tracking number. Administrators can manage customer information, courier details, delivery agents, shipments, and payment records through a centralized system. Delivery agents can view assigned deliveries and update the status of parcels during the delivery process. The main purpose of the system is to reduce manual work, improve accuracy, provide easy parcel tracking, and make courier operations faster, more organized, and efficient.

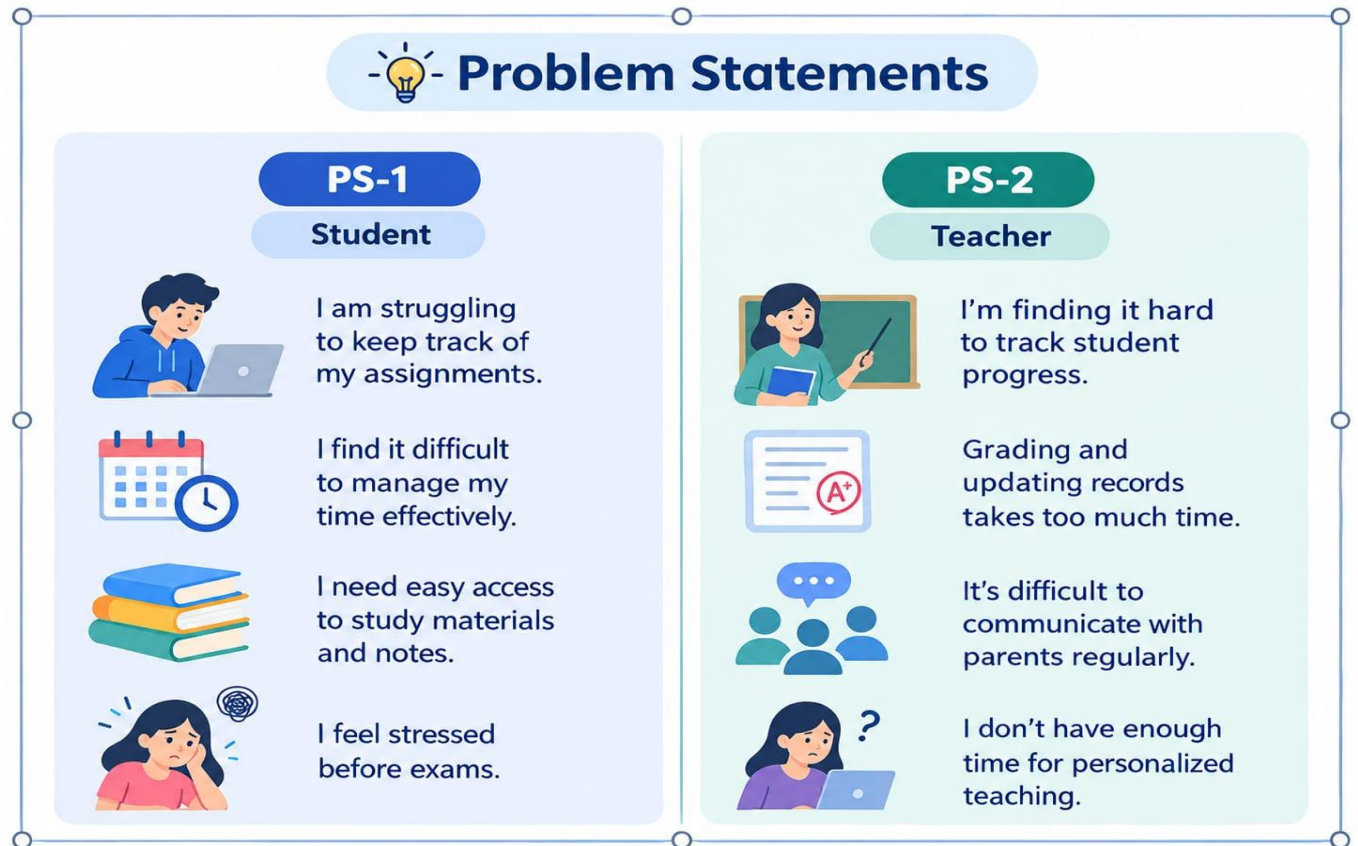
1.2 Purpose :

The purpose of the Courier Management System (CMS) is to provide an efficient and user-friendly platform for managing courier and parcel delivery operations. The system is designed to simplify courier booking, customer management, shipment tracking, delivery updates, and payment management. It helps reduce paperwork and manual errors by maintaining courier-related information in a centralized database. The system also enables customers to track their parcels and receive updated delivery information, while administrators can monitor and manage the overall courier process effectively. Overall, the purpose of the system is to improve the speed, accuracy, transparency, and efficiency of courier management.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Student	Access Study mat anytime	Resources are scattered	No single learning platform	Frustrated
PS-2	Teacher	Monitor student progress	Reports are difficult to generate	Data is stored in different systems	Overwhelmed

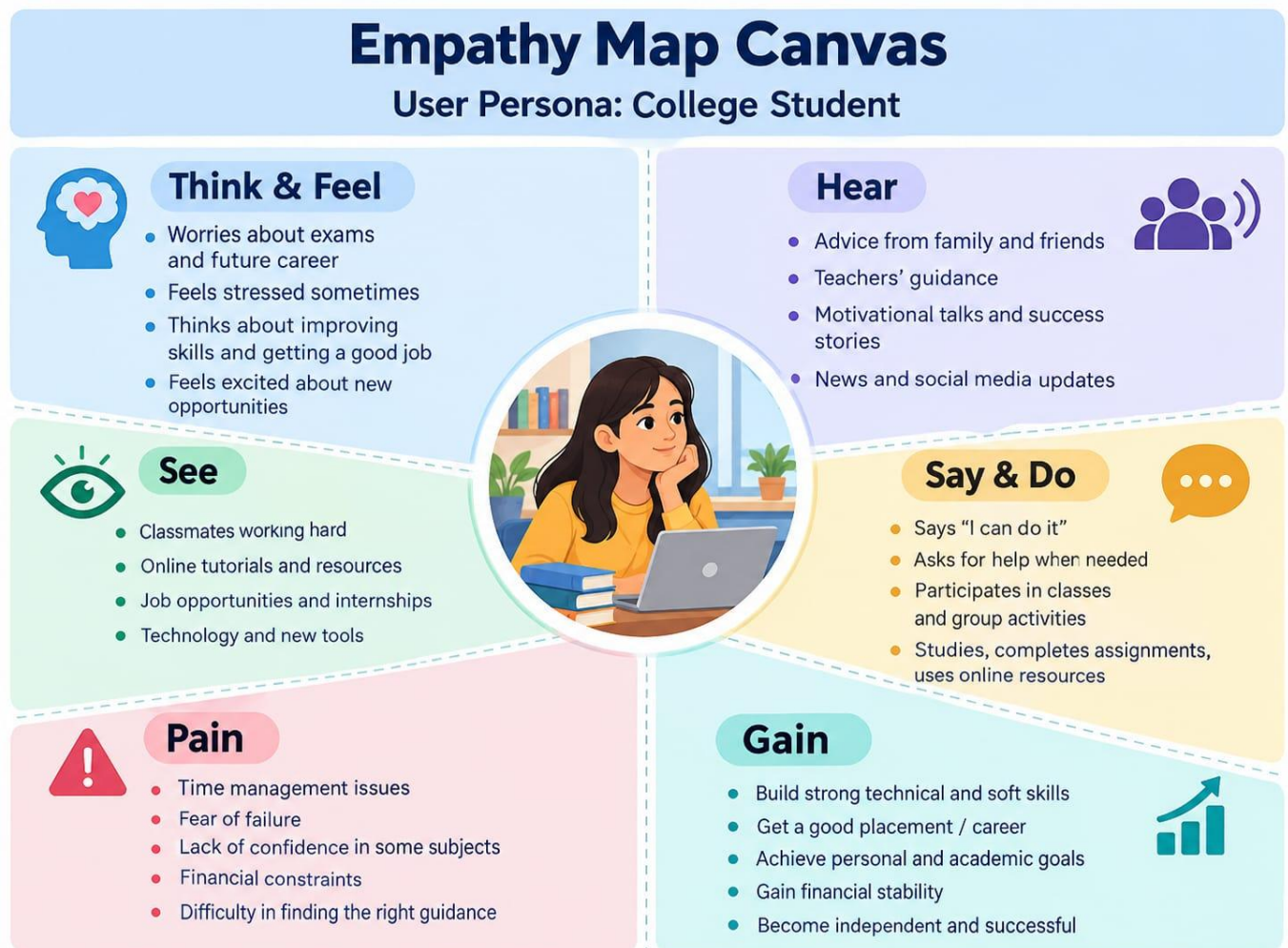


2.2 Empathy Map Canvas

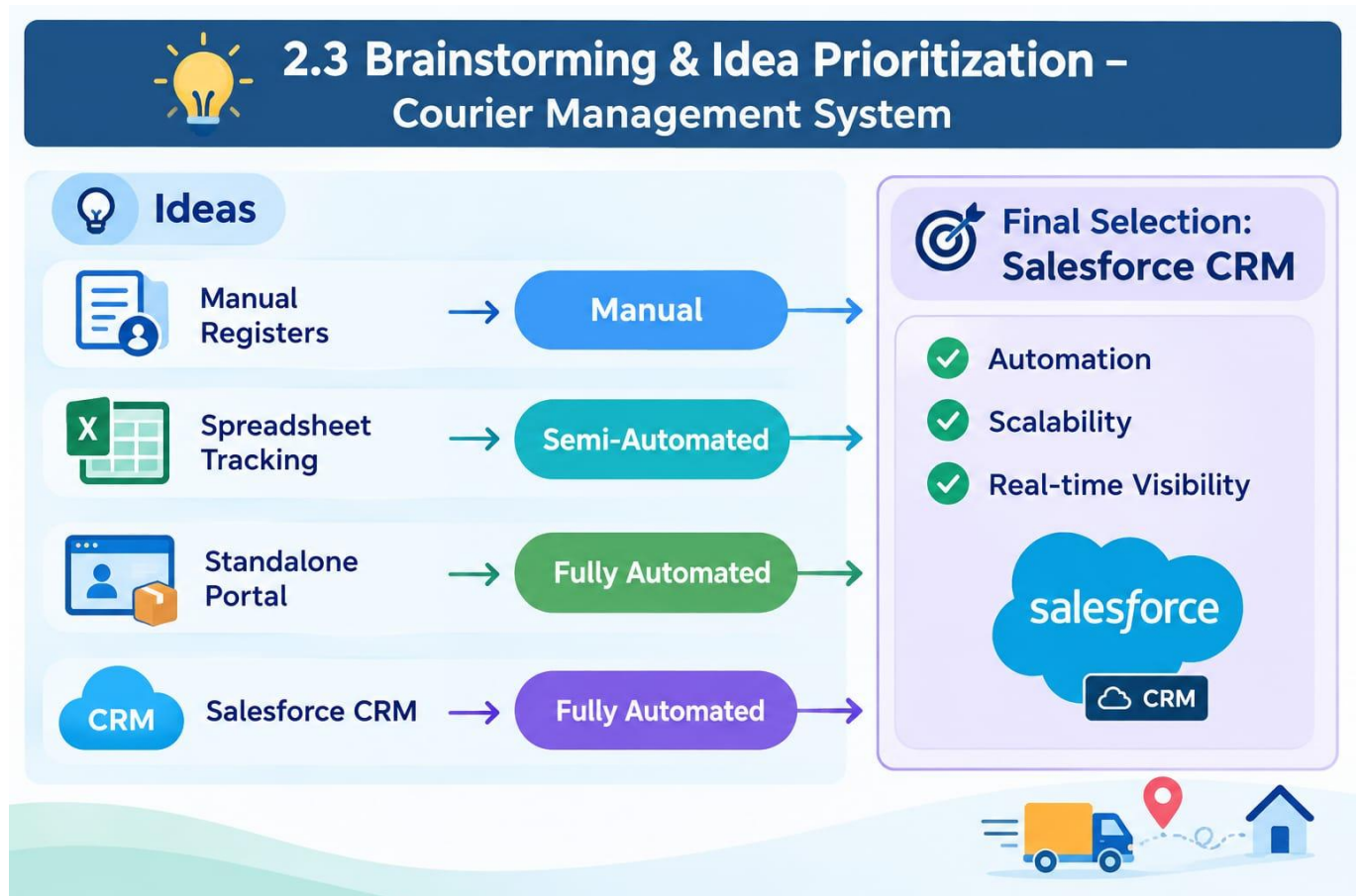
User Persona: Student

Category	Student
Think & Feel	Concerned about parcel delays and wants to know the delivery status
Hear	Updates from courier staff, friends, and family
See	Courier booking details, tracking information, and delivery status
Say & Do	Books couriers, checks tracking status, and contacts the courier service for updates

Category	Student
Pain	Delayed deliveries, lack of tracking updates, and difficulty getting shipment information
Gain	Easy booking, real-time tracking, timely notifications, and quick delivery updates



- Automation
- Scalability
- Real-time visibility



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

- Booking → Shipment Creation → Agent Assignment → Transit → Delivery
Update → Invoice → Reporting

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Registration and Login
FR-2	Customer Profile Management
FR-3	Shipment Tracking
FR-4	Delivery Status Notification
FR-5	Payment Processing

Non-Functional Requirements

NFR	Description
Usability	Simple and user-friendly interface
Security	Secure authentication and data protection

Maintainability	Easy to update and maintain the system
Compatibility	Works across different devices and browsers
Data Integrity	Ensures shipment and customer data remains accurate
Backup & Recovery	Regular data backup and quick recovery from failures

3.3 Data Flow Diagram

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery
Update → Invoice → Reports

3.4 Technology Stack

Layer	Technology
Frontend	HTML, CSS, JavaScript
Application Logic	Apex Classes
Data Validation	Validation Rules
Deployment	Salesforce Change Sets
Integration	REST API
Reporting	Salesforce Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The Courier Management System (CMS) provides a centralized solution for managing customer details, shipment bookings, delivery agent allocation, real-time tracking, payment processing, and report generation. By integrating these activities into a single Salesforce platform, the system reduces manual work, improves data accuracy, and enables efficient courier operations.



4.2 Proposed Solution

Functional Requirements

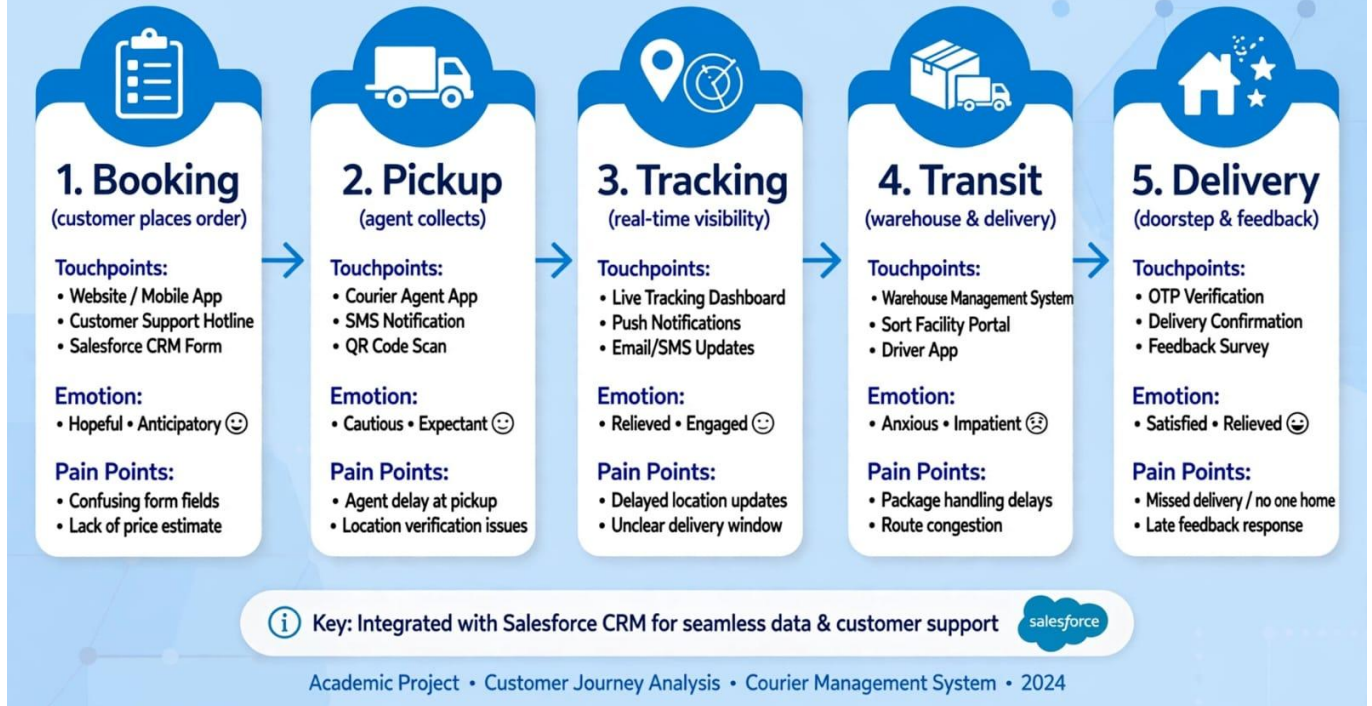
Parameter	Description
Business Model	Online courier booking and delivery services
Scalability	Supports multiple users, branches, and shipments
Customer Service	Quick access to shipment details and delivery status
Impact	Improved delivery efficiency and customer satisfaction
Innovation	Real-time shipment monitoring and automated notifications
Reporting	Automated operational and delivery reports

4.3 Solution Architecture

Components:

- User Management Module
- Shipment Management Module
- Notification Module
- Automation Workflow
- Reports
- Dashboards

2.5 Customer Journey Map - Courier Management System



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Project Configuration	USN-1	As an admin, I want to configure the Salesforce environment so that the project can be developed and managed efficiently.	3	High	Member1

Sprint-2	Customer Management	USN-2	As a staff member, I want to register and maintain customer details so that customer information is easily accessible.	5	High	Member2
----------	---------------------	-------	--	---	------	---------

Iterative Development

Requirement Prioritization

User Stories

Sprint Planning

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

The **product backlog** contains all planned features and tasks of the Courier Management System. Requirements are prioritized and divided into user stories. Each story is assigned story points based on its complexity. The selected stories are then organized into multiple sprints for systematic development and delivery.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Customer Management	USN-1	As a staff member, I want to add and update customer profiles so that customer information can be maintained accurately.	5	High	Member1
Sprint-2	Shipment Management	USN-2	As a customer, I want to create shipment details with pickup and delivery information so that my package can be registered for delivery.	5	High	Member2
Sprint-3	Automation	USN-4	As a system, I want to validate shipment details such as delivery dates, addresses, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a dispatcher, I want automated flows for shipment status updates and agent assignment so that deliveries are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for managers, agents, and staff so that sensitive shipment and billing data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a branch manager, I want reports showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a branch manager, I want dashboards showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	Medium	Member4

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
--------	--------------------	----------	-------------------	---------------------------	---	------------------------------

Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

Project Tracker, Velocity & Burndown Chart: (4 Marks)

6. Project Development Phase:

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Developing reports & dashboards
7. Setting up security with profiles, permission sets, and sharing
8. End-to-end application testing

Milestone 1: Salesforce Account

Introduction:

New to Salesforce and unsure where to start? Curious how real business applications—like loan processing and payment tracking—can be built on the platform? You're in the right place. Welcome to the Loan Management System project!

Salesforce offers powerful tools that help automate processes, improve accuracy, and manage customer operations with ease. In this project, you'll learn how Salesforce can streamline lending workflows—from borrower onboarding to loan approval, EMI tracking, and document verification.

What Is Salesforce ?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity 1: Creating Developer Account

Creating a developer org in salesforce.

1. Go to <https://developer.salesforce.com/signup> 2.

On the sign up form, enter the following details :

3. First name & Last name
4. Email

Build for the Agentic Enterprise. Sign up for your Developer Edition.

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
- ✓ Connect Claude Code, Cursor, or Codex to your org with hosted MCP
- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required
- ✓ Drag-and-drop when you want it, Apex when you need it
- ✓ 60+ MCP tools and 30+ coding skills, ready out of the box

Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data 360

First name	Last name
Tamilselvi	P
Job title	Company
Developer	Sri Akilandeswari women...
Country/region	Work email
India	tamilselvi231006@gmail...

☐ I agree to the [Main Services Agreement – Developer Services](#), [Salesforce Program Agreement](#), and [API Terms of Service](#)

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

We value your privacy. To learn more, visit our [Privacy Statement](#).

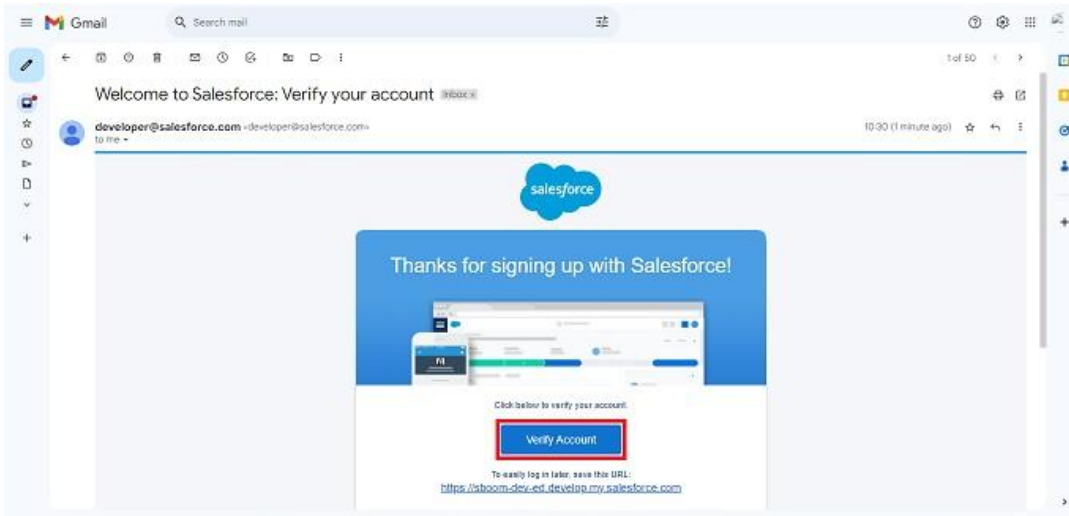
Sign me up

- 1) First name & Last name
- 2) Email
- 3) Job Title: Developer
- 4) Company: College Name
- 5) Country: India

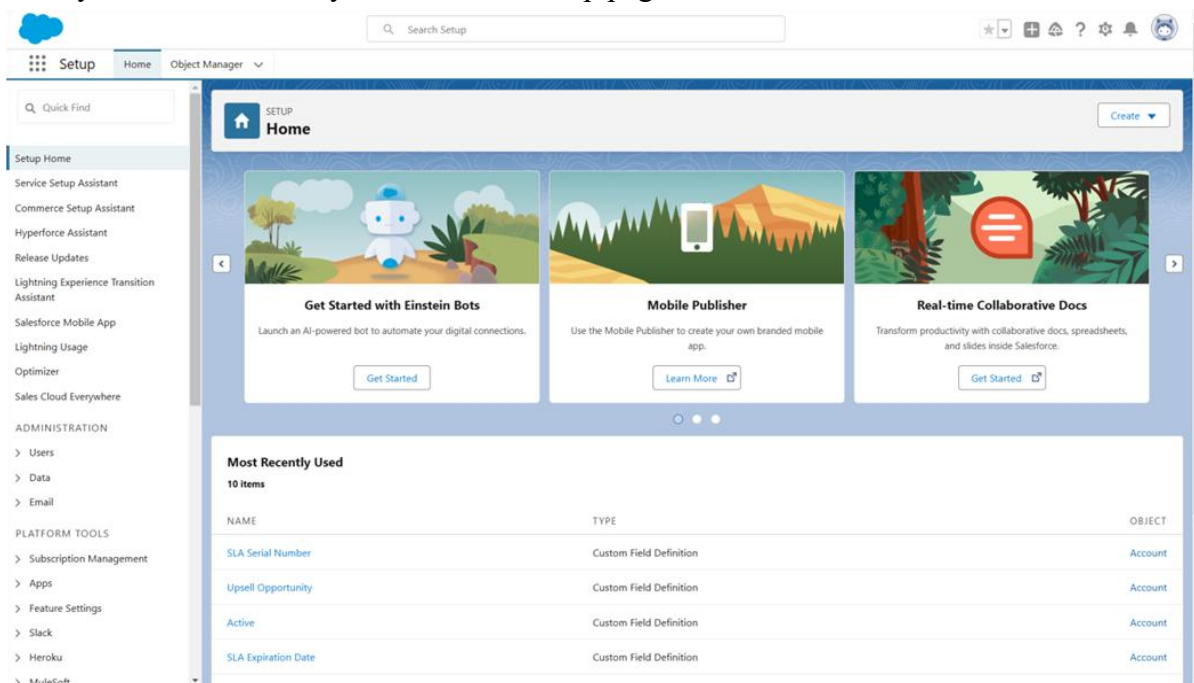
Click on sign me up after filling these.

Activity 2: Account Activation

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30 mins and sometimes 2 hours.



2. Click on Verify Account
3. Give a password and answer a security question and click on change password.
4. Then you will redirect to your salesforce setup page.



Milestone 2- Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

Custom Objects Created:

1. **Customer__c** – stores customer personal and contact information
2. **Shipment__c** – manages parcel details, source, destination, and tracking information
3. **Delivery_Agent__c** – maintains delivery agent profiles and branch details
4. **Branch__c** – represents courier branch offices
5. **Delivery_Update__c** – stores status updates for each shipment
6. **Invoice__c** – manages billing details for delivered shipments

Relationships:

Customer → Shipment (Lookup)

Shipment → Delivery_Agent (Lookup)

Delivery_Agent → Branch (Lookup)

Delivery_Update → Shipment (Master-Detail)

Invoice → Shipment (Lookup)

Steps to Implement :

Activity 1: Create Customer Object

1. Click Setup → Object Manager → Create → Custom Object
2. Enter the following details:



Search Setup



Setup

Home

Object Manager

 **Object Manager**
53+ Items, Sorted by Label

Quick Find

Schema Builder

Create

Custom Object

Custom Object from Spreadsheet

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED
Account	Account	Standard Object		

\

- Label: Customer
- Plural Label: Customers
- Record Name: Customer Name (Text)

3. Under Optional Features:

-

Enable Allow Reports

- Enable Track Field History
- In Search Status check the box Allow Search
- In Object Creation Options check the box Add Notes and Attachments related list to default page layout

4. Click Save.

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☒ Track Field History
- ☐ Allow In Chatter Groups
- ☐ Enable Licensing [1](#)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

Deployment Status

☐ In Development
☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

- ☒ Allow Search

Object Creation Options (Available only when custom object is first created)

- ☒ Add Notes and Attachments related list to default page layout
- ☐ Launch New Custom Tab Wizard after saving this custom object

[Save](#) [Save & New](#) [Cancel](#)

Activity 2: Create Shipment Object

1. Go to Setup → Object Manager → Create → Custom Object
2. Enter:
 - Label: Shipment
 - Plural Label: Shipments
 - Record Name: Tracking Number (Auto Number)
3. Example Format: SHP-{000000}
Starting Number: 0

- 4. Enable:
Allow Reports
- Track Field History

5. Click Save

Activity 3: Create Delivery_Agent Object

1. Setup → Object Manager → Create → Custom Object
2. Enter:
 - Label: Delivery Agent
 - Plural Label: Delivery Agents
 - Record Name: Agent Name (Text)
3. Enable:
 - Allow Reports
 - Track Field History

4. Click Save

Activity 4: Create Branch Object

1. Setup → Object Manager → Create → Custom Object
2. Enter:
 - Label: Branch

- Plural Label: Branches
- Record Name: Branch Name (Text)

3. Enable:

- Allow Reports

4. Click Save.

Activity 5: Create Delivery_Update Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Delivery Update
- Plural Label: Delivery Updates
- Record Name: Update Number (Auto Number)

3. Example Format: UPD-{0000}

Enter Starting number : 0

4. Enable:

- Allow Reports
- Track Field History

5. Click Save.

Activity 6: Create Invoice Object

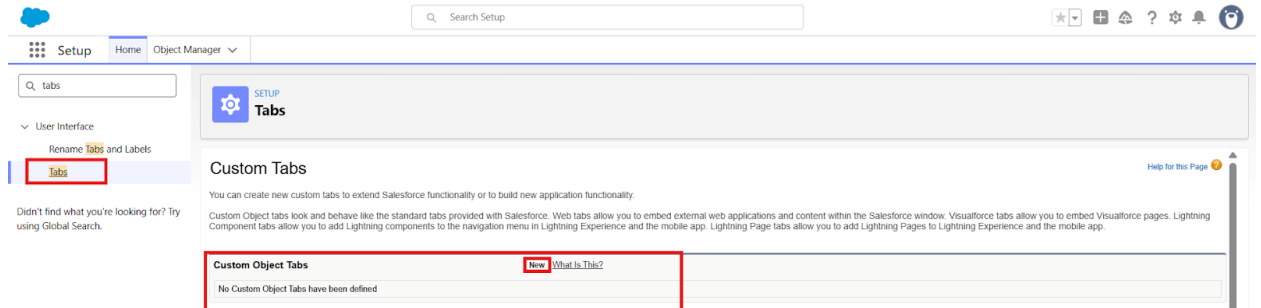
1. Setup → Object Manager → Create → Custom Object
2. Enter:
 - Label: Invoice
 - Plural Label: Invoices
 - Record Name: Invoice Number (Auto Number)
3. Example Format: INV-{0000}
Enter Starting number: 0
4. Enable:
 - Allow Reports
 - Track Field History
5. Click Save

Milestone 3 - Tabs Tab:

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1: Creating a Tab for Customer Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.



3. For Object, select Customer.
4. For Tab Style, choose any icon.
5. Leave all default settings as they are.
6. Click Next, Next, and Save.

Activity 2: Creating a Tab for Shipment Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Shipment.
4. Choose any Tab Style.
5. Keep defaults unchanged.
6. Click Next, Next, and Save.

Activity 3: Creating a Tab for Delivery Agent Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Delivery Agent.
4. Pick any Tab Style.

5. Leave default settings.
6. Click Next, Next, and Save.

Activity 4: Creating a Tab for Branch Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Branch.
4. Choose any Tab Style.
5. Keep all defaults.
6. Click Next, Next, and Save.

Activity 5: Creating a Tab for Delivery Update Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Delivery Update.
4. Select any Tab Style.
5. Keep default configurations.
6. Click Next, Next, and Save.

Activity 6: Creating a Tab for Invoice Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.

3. For Object, select Invoice.
4. Select any Tab Style.
5. Leave defaults unchanged.
6. Click Next, Next, and Save.

Milestone 4: Fields & Relationships

What are fields ?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1: Creation of fields for the Customer Object

Field Name	API Name	Data Type	Required	Description
Customer Name	Name	Text	Yes	Full name of customer
Email	Email__c	Email	Yes	Customer email
Phone	Phone__c	Phone	Yes	Customer contact number
Address	Address__c	Text Area	Yes	Customer address
Customer Type	Customer_Type__c	Picklist	Yes	Regular / Business
City	City__c	Text	No	Customer city
State	State__c	Text	No	Customer state
Pincode	Pincode__c	Text	No	Postal code

Is Active	Is_Active__c	Checkbox	No	Active status
-----------	--------------	----------	----	---------------

1. Email (Email Field)

- Go to Setup → Object Manager → Customer__c
- Click Fields & Relationships → New
- Select Email → Next
- Field Label: Email
- Check the Required box
- Click Next → Next → Save

2. Phone (Phone Field)

- Setup → Object Manager → Customer__c
- Fields & Relationships → New
- Select Phone → Next
- Field Label: Phone
- Check Required
- Click Next → Next → Save

3. Address (Text Area)

- Setup → Object Manager → Customer__c
- Fields & Relationships → New
- Select Textarea → Next
- Field Label: Address
- Check Required

- Save

4. Customer Type (Picklist)

- Setup → Customer__c → Fields → New
- Picklist → Next
- Field Label: Customer Type
- Enter values:
 - Regular
 - Business
- Check Required
- Save

5. City (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **City**
- Save

6. State (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **State**
- Save

7. Pincode (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **Pincode**
- Save

8. Is Active (Checkbox)

- Setup → Customer__c → Fields → New
- Checkbox → Next
- Field Label: **Is Active**
- Save

Activity 2: Creation of fields for the Shipment Object

Field Name	API Name	Data Type	Required	Description
Tracking Number	Name	Auto Number	Yes	Unique ID
Customer	Customer__c	Lookup(Customer)	Yes	Related customer
Source Address	Source_Address__c	Text Area	Yes	Pickup location
Destination Address	Destination_Address__c	Text Area	Yes	Delivery location

Weight	Weight__c	Number	Yes	Parcel weight
Price	Price__c	Currency	Yes	Shipping charge
Status	Status__c	Picklist	Yes	Shipment status
Expected Date	Expected_Date__c	Date	Yes	Delivery estimate
Assigned Agent	Assigned_Agent__c	Lookup(Delivery_Agent)	No	Delivery agent
Shipment Type	Shipment_Type__c	Picklist	Yes	Domestic/International

1. Source Address (Text Area)

- Setup → Shipment__c → Fields → New
- Textarea → Next
- Field Label: **Source Address**
- Required → Save

2. Destination Address (Text Area)

- Setup → Shipment__c → Fields → New
- Textarea → Next
- Field Label: **Destination Address**
- Required → Save

3. **Weight (Number)**

- Setup → Shipment__c → Fields → New
- Number → Next
- Field Label: **Weight**
- Required → Save

4. **Price (Currency)**

- Setup → Shipment__c → Fields → New
- Currency → Next
- Field Label: **Price**
- Required → Save

5. **Status (Picklist) Values:**

- Booked
- In Transit
- Out for Delivery
- Delivered
- Cancelled

Required → Save

6. **Expected Date (Date)**

- Setup → Shipment__c → Fields → New

- Date → Next
- Field Label: **Expected Date**
- Required → Save

7. Shipment Type (Picklist)

Values:

- Domestic
- International

Required → Save

Activity 3: Creation of fields for Delivery_Agent Object

Field Name	API Name	Data Type	Required
Agent Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Branch	Branch__c	Lookup	Yes
Status	Status__c	Picklist	Yes
Joining Date	Joining_Date__c	Date	No

1. Email

- Setup → Delivery_Agent__c → Fields → New

- Email → Next
- Field Label: Email
- Required → Save

2. Phone

- Setup → Delivery_Agent__c → Fields → New
- Phone → Next
- Field Label: Phone
- Required → Save

3. Status (Picklist)

Values:

- Active
- Inactive
- On Leave

Required → Save

4. Joining Date

- Setup → Delivery_Agent__c → Fields → New
- Date → Next
- Field Label: Joining Date
- Save

Activity 4: Creation of fields for Branch Object

Field Name	API Name	Data Type	Required
Branch Name	Name	Text	Yes
Branch Code	Branch_Code__c	Text	Yes
City	City__c	Text	Yes
State	State__c	Text	Yes
Manager Name	Manager_Name__c	Text	No

Create each field using:

- Setup → Branch__c → Fields → New
- Select respective Data Type
- Enter Field Label
- Save

(Repeat for all fields)

Activity 5: Creation of fields for Delivery_Update Object

Field Name	API Name	Data Type	Required
Update Number	Name	Auto Number	Yes
Shipment	Shipment__c	Master-Detail	Yes
Status	Status__c	Picklist	Yes
Update Date	Update_Date__c	DateTime	Yes
Remarks	Remarks__c	Text	No

Status Picklist Values:

- In Transit
- Out for Delivery
- Delivered
- Delayed

Required → Save

Activity 6: Creation of fields for Invoice Object

Field Name	API Name	Data Type	Required
Invoice Number	Name	Auto Number	Yes
Shipment	Shipment__c	Lookup	Yes
Amount	Amount__c	Currency	Yes
Invoice Date	Invoice_Date__c	Date	Yes
Payment Status	Payment_Status__c	Picklist	Yes

Payment Status Values

- Paid
- Unpaid
- Pending

Required → Save

Relationships :

Child Object	Field	Parent Object	Type
Shipment	Customer__c	Customer	Lookup
Shipment	Assigned_Agent__c	Delivery_Agent	Lookup
Delivery_Agent	Branch__c	Branch	Lookup
Delivery_Update	Shipment__c	Shipment	Master-Detail
Invoice	Shipment__c	Shipment	Lookup

Relationship Creation Steps :

Activity 1 – Shipment → Customer

- Setup → Shipment__c → Fields → New
- Lookup Relationship
- Related Object: **Customer__c**
- Field Label: **Customer**
- Save

Activity 2 – Shipment → Delivery Agent

- Setup → Shipment__c
- New → Lookup
- Object: **Delivery_Agent__c**
- Label: **Assigned Agent**

- Save

Activity 3 – Delivery Agent → Branch

- Setup → Delivery_Agent__c
- New → Lookup
- Related Object: **Branch__c**
- Label: **Branch**
- Save

Activity 4 – Delivery Update → Shipment

- Setup → Delivery_Update__c
- New → **Master-Detail**
- Related Object: **Shipment__c**
- Save

Activity 5 – Invoice → Shipment

- Setup → Invoice__c
- New → Lookup
- Object: **Shipment__c**
- Save

Milestone 5-Validation Rules :

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.

Activity 1: Create Validation Rules on Customer__c

1. Validation Rule: Email Must Not Be Blank

Steps:

- Click the Gear icon → Setup
- Go to Object Manager
- Search and select Customer__c
- From the left menu, click Validation Rules
- Click New
- Rule Name: Email_Required
- Check Active

In Error Condition Formula, paste:

ISBLANK(Email__c)

- Click Check Syntax
- Error Message: Email is required.
- Error Location: Field → Email
- Click Save.

2. Validation Rule: Phone Must Be 10 Digits

Steps:

- Setup → Object Manager → Customer__c
- Validation Rules → New
- Rule Name: Validate_Phone_Number
- Check Active

Formula:

$\text{LEN}(\text{Phone_c}) \leq 10$

- Check Syntax
- Error Message: **Phone number must be 10 digits.**
- Error Location: Field → Phone
- Save

3. Validation Rule: Pincode Must Be 6 Digits

Steps:

- Setup → Object Manager → Customer__c
- Validation Rules → New
- Rule Name: Validate_Pincode
- Active → Yes

Formula:

$\text{LEN}(\text{Pincode_c}) \leq 6$

- Error Message: **Pincode must be 6 digits.**
- Location: Field → Pincode
- Save

Activity 2: Create Validation Rules on Shipment__c

1. Validation Rule: Weight Must Be Greater Than Zero

Steps:

- Setup → Object Manager → Shipment__c
- Click Validation Rules → New
- Rule Name: Validate_Weight
- Active → Checked

Formula:

Weight__c <= 0

- Check Syntax

Error Message: **Weight must be greater than zero.**

- Location: Field → Weight
- Save

2. Validation Rule: Price Must Be Greater Than Zero

Steps:

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Price
- Active → Yes

Formula:

Price__c <= 0

- Error Message: **Price must be greater than zero.**
- Location: Field → Price
- Save

3. Validation Rule: Expected Date Cannot Be in the Past

Steps:

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Expected_Date

Formula:

Expected_Date__c < TODAY()

-

- Check Syntax
- Error: **Expected delivery date cannot be in the past.**
- Location: Field → Expected Date
- Save

Activity 3: Create Validation Rules on Delivery_Agent__c

1. Validation Rule: Email Must Be Valid

Steps:

- Setup → Object Manager → Delivery_Agent__c
- Validation Rules → New
- Rule Name: Validate_Agent_Email
- Active → Yes

Formula:

NOT(CONTAINS(Email__c, "@"))

- Error Message: **Enter a valid email address.**
 - Location: Field → Email
- Save

Activity 4: Create Validation Rules on Branch__c

1. Validation Rule: Branch Code Must Not Be Blank

Steps:

- Setup → Object Manager → Branch__c
- Validation Rules → New
- Rule Name: Branch_Code_Required

Formula:

ISBLANK(Branch_Code__c)

- Error Message: **Branch Code is mandatory.**
- Location: Field → Branch Code
- Save

Activity 5: Create Validation Rules on Delivery_Update__c

1. Validation Rule: Remarks Required If Status is Delayed

Steps:

- Setup → Object Manager → Delivery_Update__c
- Validation Rules → New
- Rule Name: Remarks_Required_For_Delay

Formula:

-

```
AND(  
  ISPICKVAL(Status__c, "Delayed"),  
  ISBLANK(Remarks__c)  
)
```

- Error Message: **Remarks are required when status is Delayed.**
- Location: Field → Remarks
- Save

Milestone 6 - Page Layouts :

Page Layouts in Salesforce control what a user sees on a record page and how information is organized.

They help structure fields, sections, buttons, related lists, and other components in a clean and user-friendly way.

By customizing page layouts, we ensure that users can easily enter, update, and view the required data.

Activity 1: Edit Page Layout for Customer__c

Steps:

- Setup → Object Manager → Customer__c
- Click Page Layouts
- Click the default layout (usually Customer Layout)

Sections to Add & Arrange:

- Open the default layout

In the top palette, click “Section”

- Drag the Section and drop it anywhere on the layout

A pop-up appears → Fill details:

- Section Name: “Customer Information”
- Layout:
 - 1-Column or
 - 2-Column (recommended)
- Tab-Key Order: Top-Down
- Click OK

The new section will appear on the layout.

- Drop fields inside the section (it will highlight).
- Repeat the same for each field.

Section 1: Customer Information

- Customer Name
- Customer Type
- Email
- Phone
- Is Active

Section 2: Address Details

- Address
- City

State

- Pincode
9. Edit the layout as needed
 10. Click Save.

Activity 2: Edit Page Layout for Shipment__c

Steps:

- Setup → Object Manager → Shipment__c
- Click Page Layouts
- Open the default Shipment Layout

Sections to Add:

Section 1: Shipment Details

- Tracking Number
- Customer
- Shipment Type
- Weight
- Price
- Status

Section 2: Address Information

-

- Source Address
- Destination Address

Section 3: Delivery Information

- Expected Date
- Assigned Agent
- Arrange fields properly
- Click Save

Activity 3: Edit Page Layout for Delivery_Agent__c

Steps:

- Setup → Object Manager → Delivery_Agent__c
- Page Layouts
- Open default Delivery Agent Layout

Arrange Fields:

Agent Details

- Agent Name
- Email
- Phone
- Branch
- Status

Joining Date

Additional Information

- Related Shipments
- Related Delivery Updates
- Edit → Save

Activity 4: Edit Page Layout for Branch__c

Steps:

- Setup → Object Manager → Branch__c
- Page Layouts
- Open default Branch Layout

Sections:

Branch Information

- Branch Name
- Branch Code
- City
- State

-

Manager Name

- Arrange fields
- Click Save

Activity 5: Edit Page Layout for Delivery_Update__c

Steps:

- Setup → Object Manager → Delivery_Update__c
- Page Layouts
- Open default Delivery Update Layout

Sections:

Update Information

- Update Number
- Shipment
- Status
- Update Date
- Remarks
- Arrange properly
- Click Save

Activity 6: Edit Page Layout for Invoice__c

Steps:

Setup → Object Manager → Invoice__c

- Page Layouts
- Open default Invoice Layout

Sections:

Invoice Details

- Invoice Number
- Shipment
- Amount
- Invoice Date
- Payment Status
- Arrange fields as needed
- Click Save

Milestone 7 - Flows

In Salesforce, a flow is a powerful tool that allows you to automate business processes, collect and update data, and guide users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

Activity 1: Auto-Update Shipment Status When Delivery Update is Created

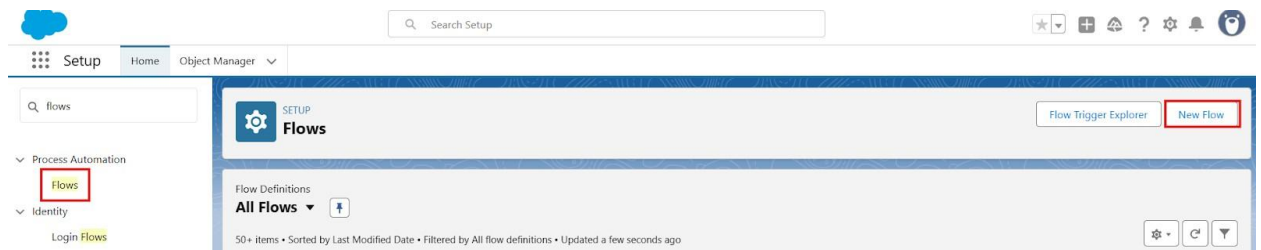
Scenario:

Whenever a Delivery_Update__c record is created, automatically update the related Shipment__c Status with the latest status from Delivery Update.

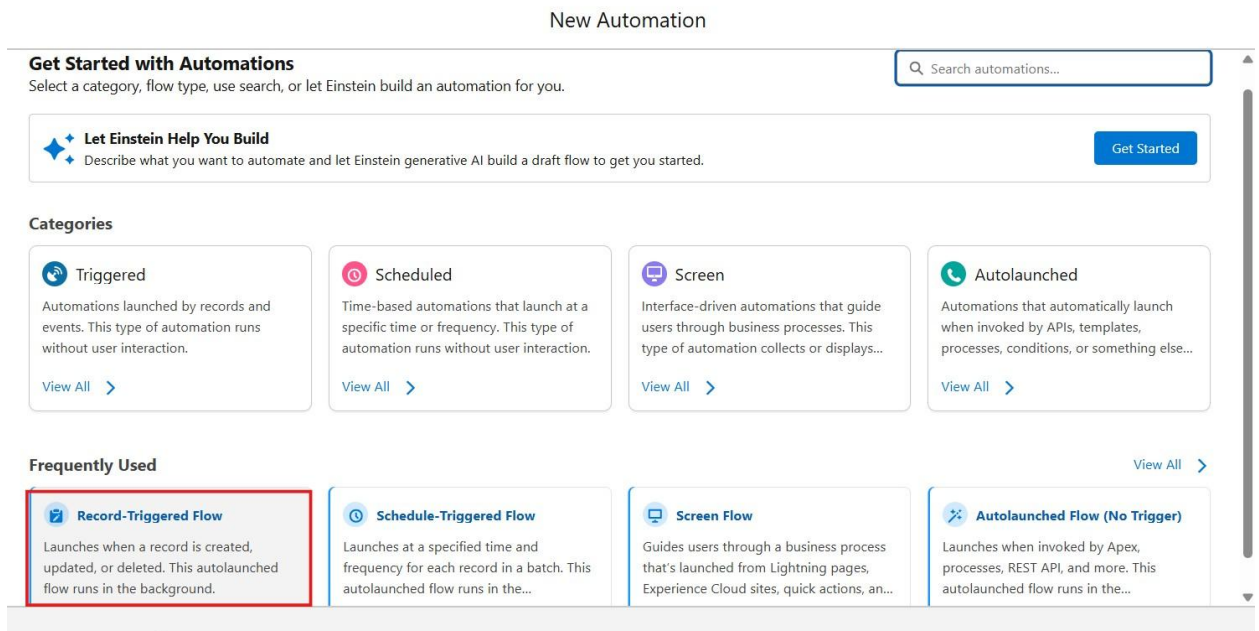
Part 1: Create the Record-Triggered Flow

Step 1: Open Flow Builder

- Click the Gear Icon → Click Setup
- In the Quick Find box (left side), type Flows
- Click Flows
- Click the New Flow button



- Select Record-Triggered Flow
- Click Create



Step 2: Configure the Start (Trigger)

On the Start panel on the right:

- Object: select Delivery_Update__c
- Trigger the Flow When:
 - Select A record is created
- Condition Requirements:
 - Choose: None (Always Run the Flow)
- Optimize the Flow For:
 - Select Actions and Related Records
- Click Done

Part 2: Update Related Shipment Record

Step 3: Add “Update Records” Element

- Click the + icon under the Start element
- Select Update Records

Configure the Update Records Element:

- Label: Update_Shipment_Status
- How to Find Records to Update:
 - Select Specify conditions to identify records

Set Filter Conditions:

- Object: Shipment__c
- Field: Id
- Operator: Equals

- Value: {!\$Record.Shipment__c}

(This means update the Shipment linked to the Delivery Update)

Set Field Values to Update Add field mapping:

Shipment Field	Value Source
Status__c	{!\$Record.Status__c}

(This copies status from Delivery Update to Shipment)

- Click Done

Part 3: Save and Activate Step

4: Save Flow

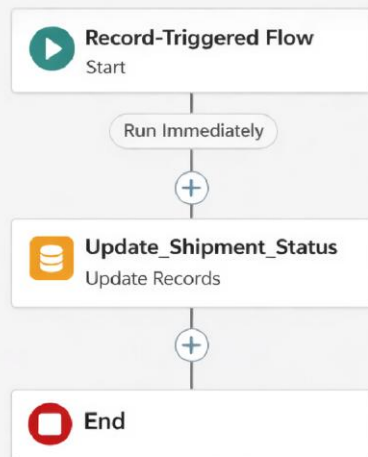
- Click Save (top-right)
- Flow Label: Auto_Update_Shipment_Status
- API Name will auto-fill
- Click Save

Step 5: Activate Flow

- Click Activate
- Flow status should show Active

Final Flow Layout:

Auto-Update Shipment Status When Delivery Update is Created



Milestone 8 - The Lightning App :

What is an App ?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1: Create a Lightning App

Create the Courier Management System App

1. From Setup, enter App Manager in the Quick Find and select App Manager.
2. Click New Lightning App.
 - Enter the Courier Management System as the App Name.
3. Upload an image or icon for the app, then click Next.

New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name ⓘ

* Developer Name ⓘ

Description ⓘ

App Branding

Image ⓘ



Clear

Primary Color Hex Value ⓘ



Org Theme Options

☐ Use the app's image and color instead of the org's custom theme

Next

4. Under App Options, leave the default selections and click Next.
5. Under Utility Items, leave as it is and click Next.
6. From Available Items, select the following and move them to Selected Items:
 - Customer
 - Shipment
 - Delivery Agent
 - Branch
 - Delivery Update
 - Invoice
 - Reports
 - Dashboards
7. Click Next.

8. From Available Profiles, select System Administrator and move it to Selected Profiles.
9. Click Save & Finish.

New Lightning App

User Profiles
Choose the user profiles that can access this app.

Available Profiles

Analytics Cloud Integration User

Analytics Cloud Security User

Anypoint Integration

Authenticated Website

Authenticated Website

B2B Reordering Portal Buyer Profile

▶

◀

Selected Profiles

System Administrator

Save & Finish

Milestone 9- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types,

Login hours & Login IP ranges.

A profile can be assigned to many users, but users can be assigned a single profile at a time.

Activity 1: Create a Custom Profile – Courier Admin

Create a Custom Profile:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. Click New Profile.

Search Setup

Setup Home Object Manager

profil

Users

Profiles

Didn't find what you're looking for? Try using Global Search.

SETUP Profiles

Profiles

All Profiles Edit | Delete | Create New View

New Profile

Action	Profile Name ↑	User License
Edit Clone	Analytics Cloud Integration User	Analytics Cloud Integration User
Edit Clone	Analytics Cloud Security User	Analytics Cloud Integration User
Edit Clone	Anypoint Integration	Identity

4. Select Existing Profile as: System Administrator.
5. Enter Profile Name: Courier Admin.
6. Click Save.

SETUP Profiles

Clone Profile

Help for this Page

Enter the name of the new profile.

You must select an existing profile to clone from. * = Required Information

Existing Profile System Administrator

User License Salesforce

Profile Name Courier Admin

You must select an existing profile to clone from.

Existing Profile System Administrator

User License Salesforce

Save Cancel

Why use System Administrator as Base?

It already has:

- Modify All Data
- Customize Application

- Full CRUD access on all objects

By cloning it, the **Courier Admin** profile automatically inherits full access.
You can later restrict permissions if required.

Activity 2: Create a Custom Profile – Branch Manager

Steps:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. Click New Profile.
4. Select Existing Profile as: Standard Platform User.
5. Enter Profile Name: Branch Manager.
6. Click Save.
7. Click Edit on the Branch Manager profile.

Set App & Tab Settings

Under Assigned Apps / Custom App Settings, set the default app to:

Courier Management System App

Ensure Visible Tabs Include:

Set these tabs as Default On or Visible:

- Customers
- Shipments
- Delivery Agents
- Branches
- Delivery Updates
- Invoices
- Reports

- Dashboards

Custom Object Permissions

Scroll to Custom Object Permissions and set access as follows:

Object	Permissions to Grant
Customer__c	Create, Read, Edit
Shipment__c	Create, Read, Edit
Delivery_Agent__c	Read, Edit
Branch__c	Read, Edit
Delivery_Update__c	Create, Read, Edit
Invoice__c	Read

Custom Object Permissions							
	Basic Access				Data Administration		
	Read	Create	Edit	Delete	View All Records 	Modify All Records 	View All Fields
Customer__c	☒	☒	☒	☒	☐	☐	☐
Shipment__c	☒	☒	☒	☒	☐	☐	☐
Delivery_Agent__c	☒	☒	☒	☐	☐	☐	☐
Branch__c	☒	☒	☒	☐	☐	☐	☐
Delivery_Update__c	☒	☒	☒	☒	☐	☐	☐
Invoice__c	☒	☐					

Session & Password Settings

Under Session Settings:

- Session Timeout: 2 Hours of Inactivity

Under Password Policies:

- User passwords expire in: Never Expires
- Minimum Password Length: 8

Click Save.

Activity 3: Create a Custom Profile – Delivery Agent

Steps:

1. Setup → Quick Find → Profiles
2. Click New Profile
3. Existing Profile: Standard Platform User
4. Profile Name: Delivery Agent
5. Click Save

App & Tab Access

Default App: Courier Management System Visible

Tabs:

- Shipments
- Delivery Updates

Object Permissions :

Object	Permissions
Shipment__c	Read, Edit
Delivery_Update__c	Create, Read, Edit
Customer__c	Read
Invoice__c	Read

Restrictions

- No access to Branch editing
- No delete permissions
- Limited reporting access

Click Save.

Activity 4: Create a Custom Profile – Customer Support**Steps:**

- Setup → Profiles → New Profile
- Base Profile: Standard User
- Profile Name: Customer Support **Access Configuration**

Tabs Visible:

- Customers
- Shipments
- Delivery Updates

Object Permissions :

Object	Access
Customer__c	Read, Edit
Shipment__c	Read, Edit
Delivery_Update__c	Read
Invoice__c	Read

Session Settings

- Session Timeout: 2 Hours
- Password Length: 8

Click Save

Final Profiles Created in Project :

Profile Name	Purpose
Courier Admin	Full system control
Branch Manager	Manage shipments and agents
Delivery Agent	Update delivery status

Customer Support	Handle customers and tracking
------------------	-------------------------------

Milestone 10- Role:

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization.

Users with higher roles have greater access to data and more control over the system.

Activity 1: Creation of Courier Admin Role

Creation of Role:

1. From the Quick Find box, search for Roles and click on the Roles option.
2. Select Set Up Roles.

Understanding Roles

Set up your Role Hierarchy to control how your organization reports on and accesses data.

Sample Role Hierarchy

View other sample Role Hierarchies: Territory-based Sample

Executive Staff
CEO President CFO VP Sales

- * View & edit data, roll up forecasts, & generate reports for all users below
- * Can't access data of other Executive Staff

Western Sales Director
Director of W. Sales

- * View & edit data, roll up forecasts, & generate reports for all users directly below
- * Can't access data of users above or at same level

Eastern Sales Director
Director of E. Sales

- * View & edit data, roll up forecasts, & generate reports for all users directly below
- * Can't access data of users above or at same level

International Sales Director
Director of Int'l Sales

- * View & edit data, roll up forecasts, & generate reports for all users directly below
- * Can't access data of users above or at same level

Western Sales Rep
CA Sales Rep OR Sales Rep

Eastern Sales Rep
NY Sales Rep MA Sales Rep

International Sales Rep
Asian Sales Rep European Sales Rep

Set Up Roles

☐ Don't show this page again

3. Click on Expand All and click Add Role under CEO.

1. Below the Branch Manager role, click on Add Role.
2. Enter Label as “Delivery Agent”.
 - Role Name will auto-populate.
3. Click Save.

Activity 4: Creating Customer Support Role Under Branch Manager

Steps:

1. Under the Branch Manager role, click on Add Role.
2. Enter Label as “Customer Support”.
3. Role Name will automatically populate.
4. Click Save.

Purpose of This Hierarchy :

- **Courier Admin** – Highest level access to all data
- **Branch Manager** – Manages branch-level operations
- **Delivery Agent** – Access only to assigned shipments
- **Customer Support** – Limited access to customer and shipment tracking

This hierarchy ensures proper data sharing and security in the Courier Management System.

Milestone 11- User :

What is a user?

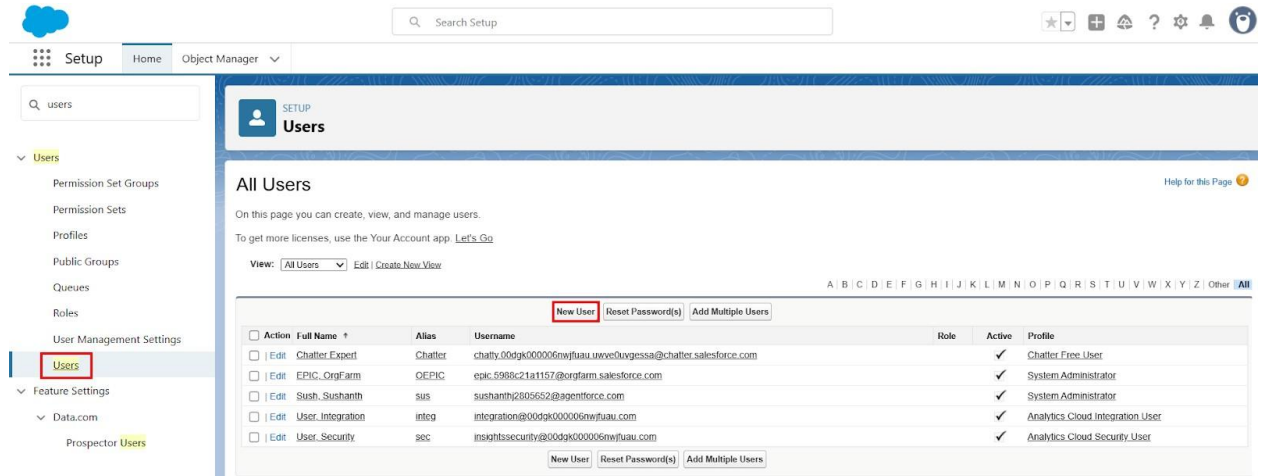
A user is anyone who logs in to Salesforce. Users are employees at your company, such as sales, reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, So If you don't find a salesforce license then deactivate a user who has a salesforce license Or change the license type from Salesforce to any other.

Activity 1: To Create a User

Steps:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



Fill in the Fields:

Field Name	Value
First Name	Enter first name (Example: Ravi)
Last Name	Enter last name (Example: Kumar)
Alias	Give an alias name
Email ID	Enter your personal email ID
Username	Must be in the format: text@text.text and must be unique
Nickname	Give a nickname
Role	Select Courier Admin
User License	Salesforce
Profile	Courier Admin

3. Click Save.

Activity 2: Creating Other Users

Repeat the same steps as **Activity 1** for creating additional users.

Use the following details:

User Type	Role	User License	Profile
Branch Manager	Branch Manager	Salesforce	Branch Manager
Delivery Agent	Delivery Agent	Salesforce Platform	Delivery Agent
Customer Support	Customer Support	Salesforce Platform	Customer Support

Example:

To create a Branch Manager User:

- Go to Setup → Users → New User
- Fill details like:

Field	Example Value
First Name	Arjun
Last Name	Reddy
Email	arjun@gmail.com
Username	arjun.manager@cms.com
Role	Branch Manager
Profile	Branch Manager

- Click Save

\

To Create Delivery Agent User:

Follow same steps with:

- Role: Delivery Agent

- Profile: Delivery Agent

To Create Customer Support User:

- Role: Customer Support
- Profile: Customer Support

Important Points

- Every user must have:
 - Unique username
 - Valid email
 - Assigned Role

- - Assigned Profile

Profiles control permissions

- Roles control data visibility

Milestone 15 : Reports & Dashboards

Reports and Dashboards help analyze business data and track courier operations efficiently.

Activity 1: Create Report Types

Step 1: Open Report Types

1. Click Gear Icon → Select Setup.
2. In Quick Find search Report Types.
3. Click New Custom Report Type.

Step 2: Create Shipment Report Type

- Primary Object: Shipment__c
- Label: Shipment Tracking Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Step 3: Create Customer Report Type

- Primary Object: Customer__c
- Label: Customer Summary Report
- Category: Reports

-

-

Step

-

Deployment Status: Deployed

Click Next → Save

4: Create Agent Performance Report Type

Primary Object: Delivery_Agent__c

- Label: Agent Performance Report

- Category: Reports

- Deployment Status: Deployed

- Click Next → Save **Activity**

2: Create Reports

Report 1: Shipment Status Summary Report

Step 1: Create the Report

1. Click App Launcher (9 dots) → Search Reports
2. Click New Report
3. Choose Shipment Tracking Report
4. Click Continue **Step 2: Add Columns**

Add these fields:

- Tracking Number

-

-

Step

-

- Customer

- Status

Expected Date

Assigned Agent

3: Add Filters

Show Me: All Shipments

- Date Filter: This Month

- Status Filter: Select specific Status (Optional)

Step 4: Group & Summarize

- Group rows by Status__c

- Add summary on Price

Step 5: Save Report

- Click Save & Run

- Name: Shipment Status Summary

- Folder: CMS Reports

Report 2: Agent Performance Report

-

-

Step

-

Step 1: Create Report

- New Report → Choose Agent Performance Report
- Click Continue

Step 2: Add Columns

- Agent Name
- Branch

Number of Shipments

Status

3: Group Data

Group by Delivery_Agent__c

Step 4: Save Report

- Name: Agent Performance Report
- Folder: CMS Reports
- Save & Run

Report 3: Revenue Report

Steps

- Create report using Shipment Tracking Report Type

-

-

Step

-

Add fields:

- Tracking Number
- Customer
- Price
- Invoice Status

Group by:

- Branch
- Month

Save as: Revenue by Branch Report

Activity 3: Create Dashboard

Step 1: Open Dashboards

1. Click App Launcher (9 dots)
2. Search Dashboards
3. Click New Dashboard

Step 2: Enter Dashboard Details

- Name: Courier Management Dashboard
- Folder: CMS Reports
- Click Create

Activity 4: Add Dashboard Components

Component 1: Shipment Status Overview (Donut Chart)

1. Click + Component
2. Choose report: Shipment Status Summary
3. Visual Type: Donut Chart
4. Click Add

Component 2: Agent Performance (Bar Chart)

- Select report: Agent Performance Report
- Chart Type: Bar Chart

- Click Add

Component 3: Revenue by Branch (Column Chart)

- Choose report: Revenue by Branch Report
- Chart Type: Column Chart
- Click Add

Activity 5: Save & Activate Dashboard

1. Click Save
2. Click Done

Share Dashboard With:

- Courier Admin
- Branch Managers
- Customer Support Team

Final Layout :



7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. All scenarios were validated to ensure correct performance and reliability.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows

- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9. Conclusion:

The Courier Management System successfully digitizes courier operations using Salesforce automation, improving efficiency, accuracy, and customer satisfaction.

10. FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You

